UNIT 4: ABSTRACT ALGEBRAIC SYSTEMS & LATTICES

Comprehensive Examination Notes, Solved Problems & Theorems

1. ALGEBRAIC SYSTEMS

1.1 Binary Operation

Let A be a non-empty set. A binary operation on A is an operation which combines any two elements of A and produces another element of A:

$$* : A \times A \to A$$

Thus, for every $a, b \in A$, $a * b \in A$ (Closure Property).

- **Addition on integers:** $+ : Z \times Z \to Z$. For any $a, b \in Z$, $a + b \in Z$. It is a binary operation.
- **Multiplication on natural numbers:** $\times : N \times N \to N$ is a binary operation.
- **Subtraction on natural numbers:** Not a binary operation because $2 - 5 = -3 \notin N$.

1.2 Properties of Binary Operations

1. **Closure:** An operation $*$ on A is closed if $a * b \in A$ for all $a, b \in A$.
2. **Associative:** $(a * b) * c = a * (b * c)$ for all $a, b, c \in A$.
3. **Commutative:** $a * b = b * a$ for all $a, b \in A$.
4. **Identity Element:** An element $e \in A$ is an identity element if $a * e = e * a = a$ for all $a \in A$.
    - For addition on Z: $e = 0$.
    - For multiplication on Z: $e = 1$.
5. **Inverse Element:** If e is the identity, b is the inverse of a if $a * b = b * a = e$.
    - Additive inverse: $-a$.
    - Multiplicative inverse ($a \neq 0$): $1/a$.

1.3 Algebraic System

An algebraic system is a non-empty set together with one or more operations defined on it: $(A, *)$ or $(A, +, \times)$.

Solved Problems: Algebraic Systems

> **Problem 1:** Determine whether subtraction is a binary operation on Z.
> **Solution:** Let $a, b \in Z$. Since the difference of any two integers is an integer, $a - b \in Z$.
> **Answer:** Yes, subtraction is a binary operation on Z.

> **Problem 2:** Determine whether division is a binary operation on Z.
> **Solution:** For $2, 3 \in Z$, $2 \div 3 = \frac{2}{3} \notin Z$. Closure fails.
> **Answer:** Division is not a binary operation on Z.

Problem 3: Show that multiplication is associative on Z.
Solution: $(a \times b) \times c = (ab)c = abc$ and $a \times (b \times c) = a(bc) = abc$. Hence, $(a \times b) \times c = a \times (b \times c)$.

Problem 4: Find the identity element for $a * b = a + b - 5$ on Z.
Solution: $a * e = a \implies a + e - 5 = a \implies e = 5$. Checking: $a * 5 = a + 5 - 5 = a$. Identity is $e = 5$.

Problem 5: Determine whether $a * b = a + b$ on Z is commutative and associative.
Solution: Commutative: $a + b = b + a \implies a * b = b * a$. Associative: $(a + b) + c = a + (b + c)$. Closed: Yes, Associative: Yes, Commutative: Yes, Identity: 0.

2. HOMOMORPHISM AND ISOMORPHISM

A mapping $f : (A, *) \to (B, \circ)$ is a **homomorphism** if it preserves the operation:

$$f(a * b) = f(a) \circ f(b) \quad \forall a, b \in A$$

An **isomorphism** is a bijective (one-to-one and onto) homomorphism. We write $A \cong B$.

Feature	Homomorphism	Isomorphism
Operation preserving	Yes: $f(a * b) = f(a) \circ f(b)$	Yes: $f(a * b) = f(a) \circ f(b)$
One-to-one	Need not be	Must be
Onto	Need not be	Must be
Structure	Preserved (may lose info)	Preserves complete structure

Solved Problems: Morphisms

Problem 1: Determine whether $f(x) = 3x$ from $(Z, +)$ to $(Z, +)$ is a homomorphism.
Solution: $f(a + b) = 3(a + b) = 3a + 3b = f(a) + f(b)$. Hence, f is a homomorphism.

Problem 2: Determine whether $f(x) = x + 1$ on $(Z, +)$ is a homomorphism.
Solution: $f(a + b) = a + b + 1$, but $f(a) + f(b) = (a + 1) + (b + 1) = a + b + 2$. Since $a + b + 1 \neq a + b + 2$, f is not a homomorphism.

Problem 3: Show that $f(x) = 2x$ on $(Z, +)$ is not an isomorphism.
Solution: While it is a homomorphism, for $y = 1 \in Z$, $2x = 1 \implies x = 1/2 \notin Z$. Thus it is not onto, so it cannot be an isomorphism.

Problem 4: Show that $f(x) = -x$ on $(Z, +)$ is an isomorphism.
Solution: Homomorphism: $f(a + b) = -(a + b) = -a - b = f(a) + f(b)$. One-to-one: $-a = -b \implies a = b$. Onto: For $y \in Z$, $f(-y) = y$. Thus, f is an isomorphism.

Problem 5: Let $f : Z \to Z_3$ be $f(x) = x \bmod 3$. Show f is a homomorphism but not an isomorphism.
Solution: $f(a + b) = (a + b) \bmod 3 = (a \bmod 3 + b \bmod 3) \bmod 3 = f(a) +_3 f(b)$. It is onto but not one-to-one since $f(0) = f(3) = 0$.

3. SEMIGROUPS, MONOIDS, AND GROUPS

Semigroup (Closure + Associativity) → Monoid (+ Identity) → Group (+ Inverse) → Abelian Group (+ Commutativity)

Property	Semigroup	Monoid	Group	Abelian Group
Closure	✓	✓	✓	✓
Associativity	✓	✓	✓	✓
Identity	Not required	✓	✓	✓
Inverse	Not required	Not required	✓	✓
Commutativity	Not required	Not required	Not required	✓
Standard Example	$(N, +)$	$(N_0, +)$, $(N, \times)$	$(Z, +)$, $(R^*, \times)$	$(Z, +)$, $(Z_n, +_n)$

Key Distinctions and Examples

- $(Z, -)$ is **not** a semigroup because $(5 - 3) - 1 = 1 \neq 5 - (3 - 1) = 3$.
- Positive integers $(Z^+, +)$ form a semigroup, but **not** a monoid because the identity $0 \notin Z^+$.
- 2×2 real matrices under multiplication form a **monoid** with identity $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, but not a group (singular matrices lack inverses).
- $(N, +)$ is **not** a group because inverses do not exist in N.

4. SUBGROUPS

A non-empty subset $H \subseteq G$ is a **subgroup** $(H \leq G)$ if H forms a group under the same operation of G.

- **Multiplicative Subgroup Test:** $a, b \in H \implies ab^{-1} \in H$.
- **Additive Subgroup Test:** $a, b \in H \implies a - b \in H$.

Example: Even integers $2Z \leq (Z, +)$.
For $a = 2m, b = 2n \in 2Z$, $a - b = 2(m - n) \in 2Z$. Hence, $2Z$ is a subgroup.

5. LATTICES

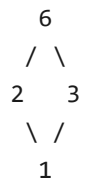
5.1 Poset and Lattice Definition

A relation $\leq$ on P is a **partial order** if it is Reflexive ($a \leq a$), Antisymmetric ($a \leq b \wedge b \leq a \implies a = b$), and Transitive ($a \leq b \wedge b \leq c \implies a \leq c$).

A poset $(L, \leq)$ is a **lattice** if every pair of elements $a, b \in L$ has:

- A greatest lower bound (GLB), called the **meet**: $a \wedge b$.
- A least upper bound (LUB), called the **join**: $a \vee b$.

5.2 Hasse Diagram Example: $L = \{1, 2, 3, 6\}$ under Divisibility



For elements 2 and 3:

$$\text{GLB: } 2 \wedge 3 = \gcd(2, 3) = 1 \qquad \text{LUB: } 2 \vee 3 = \text{lcm}(2, 3) = 6$$

5.3 Basic Laws of Lattices

- **Idempotent:** $a \vee a = a$, $a \wedge a = a$
- **Commutative:** $a \vee b = b \vee a$, $a \wedge b = b \wedge a$
- **Associative:** $(a \vee b) \vee c = a \vee (b \vee c)$, $(a \wedge b) \wedge c = a \wedge (b \wedge c)$
- **Absorption:** $a \vee (a \wedge b) = a$, $a \wedge (a \vee b) = a$

5.4 Types of Lattices

- **Bounded Lattice:** Contains a least element 0 and greatest element 1 such that $0 \leq a \leq 1, \forall a \in L$.
- **Complemented Lattice:** A bounded lattice where for every a, there exists a' such that $a \vee a' = 1$ and $a \wedge a' = 0$.
- **Distributive Lattice:** Satisfies $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$ and $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$.
- **Complete Lattice:** Every subset has both a GLB and an LUB.

6. IMPORTANT THEOREMS & PROOFS

Theorem 1: Uniqueness of Identity
Suppose e and e' are both identity elements in G.
Since e is an identity, $e * e' = e'$. Since e' is an identity, $e * e' = e$.
Therefore, $e = e'$. The identity is unique. ■

Theorem 2: Uniqueness of Inverse
Suppose b and c are both inverses of a. Then $a * b = b * a = e$ and $a * c = c * a = e$.
$b = b * e = b * (a * c) = (b * a) * c = e * c = c$.
Therefore, $b = c$. The inverse is unique. ■

Theorem 3: Inverse of a Product (Reversal Rule)
$(ab)(b^{-1}a^{-1}) = a(bb^{-1})a^{-1} = a(e)a^{-1} = aa^{-1} = e$.
Similarly, $(b^{-1}a^{-1})(ab) = e$.
Therefore, $(ab)^{-1} = b^{-1}a^{-1}$. ■

Theorem 4: Subgroup of an Abelian Group is Abelian
Let $H \leq G$ where G is abelian. For any $a, b \in H$, since $H \subseteq G$, $a, b \in G$. Since G is abelian, $ab = ba$. Therefore H is abelian. ■

7. EXAM-ORIENTED SHORT QUESTIONS (2 MARKS)

1. **Define binary operation:** A function $* : A \times A \to A$ that assigns an element of A to every ordered pair in $A \times A$.
2. **What is a semigroup?** A non-empty set with an associative binary operation.
3. **What is a monoid?** A semigroup containing an identity element.
4. **What is a group?** A monoid in which every element has an inverse.
5. **What is an abelian group?** A group G satisfying $ab = ba$ for all $a, b \in G$.
6. **Define subgroup:** A non-empty subset H of group G that itself forms a group under the operation of G.
7. **What is homomorphism?** An operation-preserving map: $f(a * b) = f(a) \circ f(b)$.
8. **What is isomorphism?** A bijective homomorphism.
9. **Define lattice:** A poset in which every pair of elements has a greatest lower bound and a least upper bound.
10. **What is meet and join?** The GLB of two elements is the meet ($a \wedge b$); the LUB is the join ($a \vee b$).